HERIOT-WATT UNIVERSITY

F79MB Statistical Models B

Assessment Project 1 – 2024: Sample solution

1. (a) Numerical summary of the work travel expenses for an IT company in Washington (in US Dollars) for years 2020 to 2023:

```
Min.   1st Qu.  Median    Mean    3rd Qu.   Max.
476     882      1056      1088     1254     2360
```

Sample standard deviation is `$319.87`.

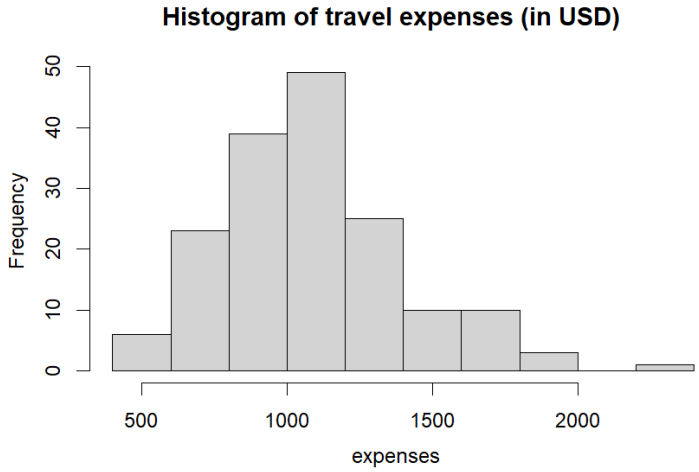


Figure 1: Histogram of the work travel expenses for an IT company in Washington (in US Dollars) for years 2020 to 2023 for a sample of 166 employees.

Comment: From the histogram (Figure 1), the distribution is unimodal and skewed to the right, with a notable outlier at $2360. Numerical summaries also indicate a right-skewed distribution, as the mean is larger than the median. The minimum travel expenses are $476 and the standard deviation is $319.87.

(b) The original data set does not appear Normally distributed — QQ plot (Figure 2(i)) is concave up, indicating right skew, also with some evidence of long right tail. The QQ plot for the square root transformed data (Figure 2(ii)) looks much closer to a normal distribution; however, some points at the upper tail lie above the straight line (qqline), showing a slight concave up, indicating a slight right skew. Similarly, the QQ plot for the log-transformed data (Figure 2(iii)) also appears much closer to a normal distribution compared to the original data. However, some points at the lower tail of the QQ plot lie below the straight line (qqline), indicating a small degree of left skew in the log-transformed data. Histograms in Figure 1 and Figure 3 further support that the data does not look normal for the original data, is slightly right-skewed for the square root transformed data, and may be slightly left-skewed for the log-transformed data.

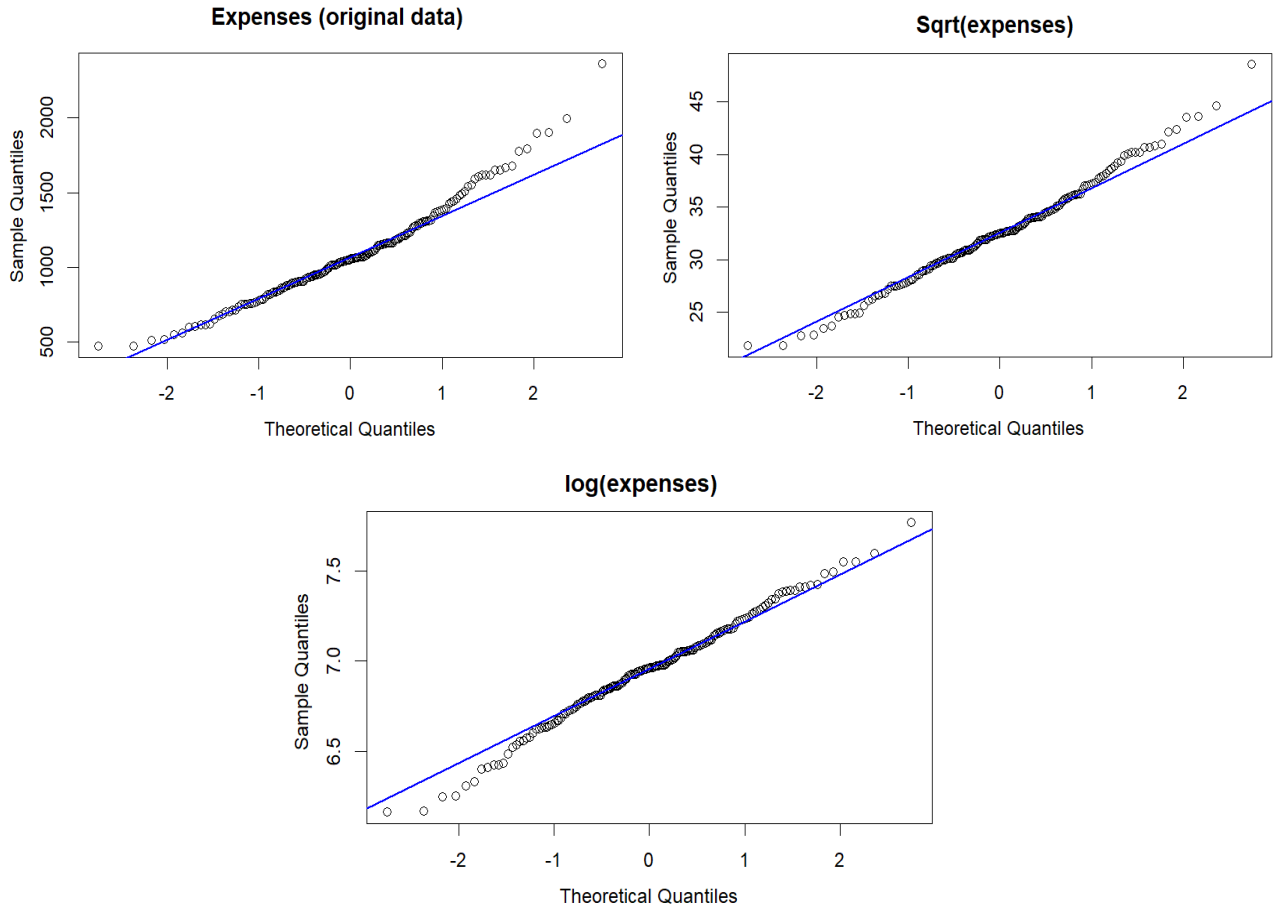


Figure 2: Normal QQ plots for (i) original data (top left); (ii) square root transformed data (top right); (iii) log-transformed data.

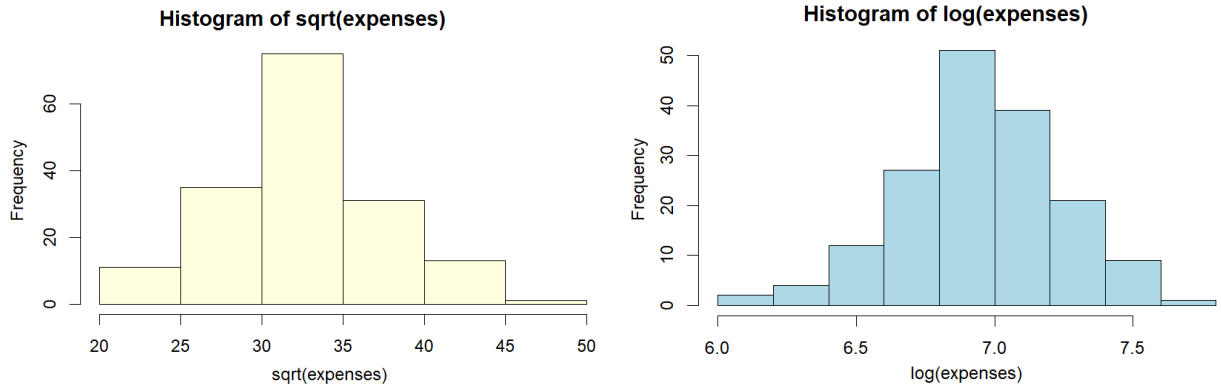


Figure 3: Histograms for (i) square root transformed data (top left); (ii) log-transformed data (top right).

(c) Perform a χ^2 goodness-of-fit

H_0: The square root transformed travel expenses data come from a Normal distribution.
H_1: These square root transformed data do not come from a Normal distribution.

Let the square root transformed data be $x_1, x_2, \cdots, x_n$. The null hypothesis does not specify the parameters $\hat{\mu}$ and $\hat{\sigma}$ of a normal distribution, so we estimate from the square root transformed data

using the typical estimators, $\hat{\mu} = \bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} = 32.63302$, $\hat{\sigma} = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}} = 4.772583$. A χ^2 test with 7 equal-probability cells, with $\hat{\mu} = 32.63302$ and $\hat{\sigma} = 4.772583$. will give all expected frequencies $e_i = \frac{166}{7} = 23.71429$, $i = 1, 2, \ldots, 7$. The equal-probability intervals (computed in *R*), together with corresponding observed frequencies, are:

Interval (sqrt(expenses))	**Observed frequency, f_i**
$(-\infty, 27.5)$	22
$[27.5, 29.9)$	22
$[29.9, 31.8)$	24
$[\ 31.8, 33.5)$	33
$[33.5, 35.3)$	23
$[35.3, 37.7)$	18
$[37.7, \infty)$	24

The test statistic value is

$$\chi^2 = \sum_{i=1}^{7} \frac{(f_i - e_i)^2}{e_i} = 5.289157.$$

With 7 cells and two estimated parameters, the test statistic χ^2 distribution has *df* = 7 –2–1 = 4, so the *p*-value = $P(\chi_4^2 > 5.289157) = 0.2588937$. There is no evidence to reject H_0. Overall, it seems that the square root transformed data could be modelled by a Normal distribution. However, should also note that there is some evidence (from QQ plot Figure 2(ii)) that the Normal distribution, while appropriate for the bulk of the square root transformed data, does not fit very well in the upper tail.

Note: Allow any other sensible way of approximating μ and σ. For instance, one alternative approach is estimating the parameters from the straight line fitted by a qqline passing through the first and the third quartiles points. In this case, $\hat{\mu} = 32.5585$ and $\hat{\sigma} = 4.240343$. The expected frequency for all cells is same as before, $e_i = \frac{166}{7} = 23.71429$, $i = 1, 2, \ldots, 7$. The equal-probability intervals (computed in *R*), together with corresponding observed frequencies, are:

Interval (sqrt(expenses))	**Observed frequency, f_i**
$(-\infty, 28)$	27
$[28, 30.2)$	24
$[30.2, 31.8)$	17
$[\ 31.8, 33.3)$	32
$[33.3, 35)$	21
$[35, 37.1)$	17
$[37.1, \infty)$	28

The test statistic value is $\chi^2 = \sum_{i=1}^{7} \frac{(f_i - e_i)^2}{e_i} = 8.240964$ with a *p*-value = $P(\chi_4^2 > 8.240964) = 0.08313967$. Therefore, there is some evidence, though not strong, to conclude that the square root transformed data is not well-modelled by a Normal distribution at the 10% level. However, it is not statistically significant at the 5% level. This result aligns with observations from the Normal QQ

plot in Figure 2(ii) and Figure 4, where some points obviously deviate from the upper tail of the straight line given by a qqline, compared to the line formed by the typical estimator.

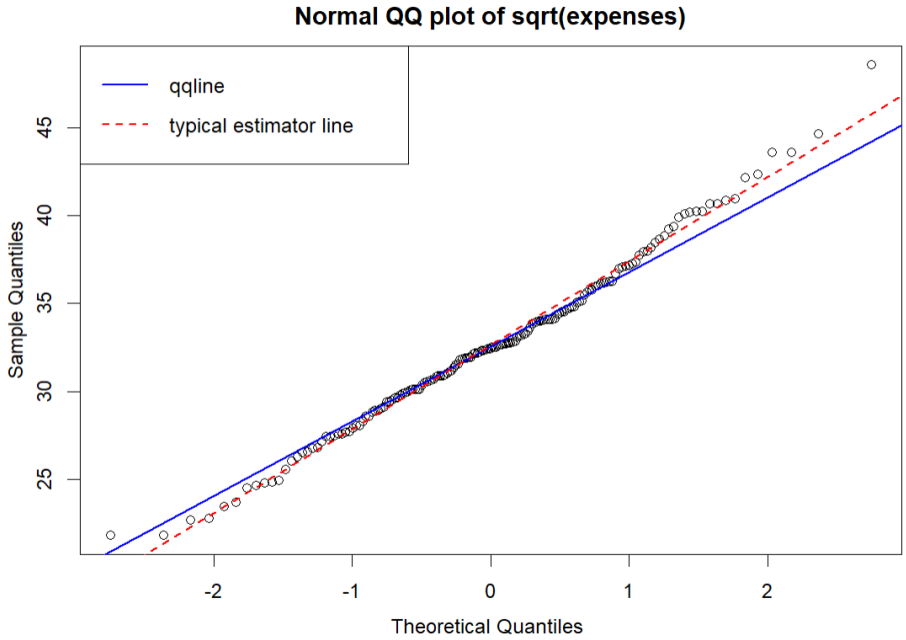


Figure 4: QQ plot for square root transformed data using qqline and typical estimator line.

(d) Sensible comments are acceptable if the correct concepts are applied in answering the questions. For example,

Case 1: If the parameters are estimated using the typical estimators

From the QQ plot in Figure 2(ii), there appear to be no noticeable outliers in the square root transformed data, although some points at the upper tail of the QQ plot lie above the straight line. This observation is further supported by the insignificant χ^2 goodness-of-fit test, with a p-value of 0.2588937, indicating that the square root-transformed data follows a Normal distribution without outliers.

Case 2: If the parameters are estimated from a qqline passing through the first and the third quantiles points

From the QQ plot in Figure 2(ii), although some data deviate from the straight line in the upper tail of the distribution, most of the points closely align with the straight line in the plot. Thus, there appear to be no outliers in the data. The χ^2 goodness-of-fit test, with a p-value of 0.08313967, indicates that we cannot conclusively state that the data comes from a Normal distribution at the 10% level. However, we do not reject the null hypothesis at the 5% significance level. Therefore, this also supports the claim that there are no potential outliers in the square root transformed data.

Comments on the implications of outliers in an analysis, possible reasons for their existence in the data, and methods for handling such outliers:

Outliers are data points that significantly differ from other observations and may represent variability in the measurement or indicate experimental errors. They can have a substantial impact on statistical analyses, affecting the mean and standard deviation of the dataset and leading to misrepresentations in the assessment of central tendency and variability. In the context of a χ^2

goodness-of-fit test, outliers can inflate the test statistic, potentially resulting in the erroneous rejection of the null hypothesis.

Possible reasons for outliers could be data entry errors or mis-recording of travel expenses. It could also be unusual events or exceptional circumstances that led to atypical expense amounts or changes in travel policy or unexpected travel requirements.

So, it is important to examine the outliers to determine if they are due to errors or genuine extreme values. Applying a transformation, such as a logarithmic or square root transformation, can be considered to reduce the impact of outliers. Besides, we can also replace outliers with imputed values, such as the mean, median, or a more sophisticated imputation method based on the characteristics of the data set.

(e) Let m the population median of the square root transformed data. Under the asymptotic theory, we should have $\hat{m} \sim N(m, \left(\frac{1}{2f(m)\sqrt{n}}\right)^2)$, where $f(\hat{m}) = \frac{1}{\sqrt{2\pi}s}$ is the density function of a Normal distribution. Given sample median $\hat{m}$=32.48841, $s = \hat{\sigma}$ =4.772583, and sample size n =166, the estimate of the standard error of median is approximately given by

$$s.e.(\hat{m}) = \frac{1}{2f(\hat{m})\sqrt{n}} = s\sqrt{\frac{\pi}{2n}} = 0.464258.$$

Then a 95% confidence interval (CI) of the population median, m, is

$$\hat{m} \pm Z_{0.025} \times s.e.(\hat{m})$$

i.e . $(31.57849,\ 33.39834)$

Next, the sample mean of the square root transformed data is $\bar{x}$ =32.63302. A 95% confidence interval (CI) of the population mean, μ, is

$$\bar{x} \pm Z_{0.025} \times \frac{s}{\sqrt{n}}$$

i.e . $(31.907,\ 33.359)$

The 95% confidence intervals obtained using asymptotic theory for the population median and the population mean of the square root travel expenses are considered very similar. This similarity arises because the square root-transformed data approximates a Normal distribution, evident in the QQ plot of Figure 2(ii), histogram of Figure 3(i) and a χ^2 goodness-of-fit test with a p-value of 0.2588937. With a sample size of n = 166, which is quite large, one would expect the asymptotic theory to perform well for the population mean. Comparing the two confidence intervals, the one based on the population mean is computationally simpler and, thus, preferred. Moreover, as the square root transformed data exhibits no skewness, justifying the use of the mean over the median is both rational and preferable. Another advantage of the CI for the mean is that it is narrower than the CI for the median, by a factor $\sqrt{\frac{\pi}{2}}$.

(f) In conclusion, the travel expenses dataset exhibits a positively skewed distribution, with a mean of \$1088, a median of \$1056, a minimum of \$476, a maximum of \$2360 (potentially an outlier), and a standard deviation of \$319.87. From the QQ plot and the chi-squared goodness-of-fit tests, the square root transformed data appears reasonably modelled using a Normal distribution, with no outliers identified. When we perform 95% confidence intervals based on asymptotic theory for the

population median and the population mean of the square root travel expenses, the intervals are very close. Of the two intervals, the one for the population mean is preferred because it is easier to compute, and the square root transformed data approximates closer to a Normal distribution. Additionally, the sample size is large, justifying the suitability of finding the confidence interval for the population mean using the Central Limit Theorem.

2. (a) Numerical summary of the BMW used car prices in the United Kingdom (in £) for year 2023 of a sample of 28 data set:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
9400	12712	15000	17121	17875	36995

Sample standard deviation is £7043.108.

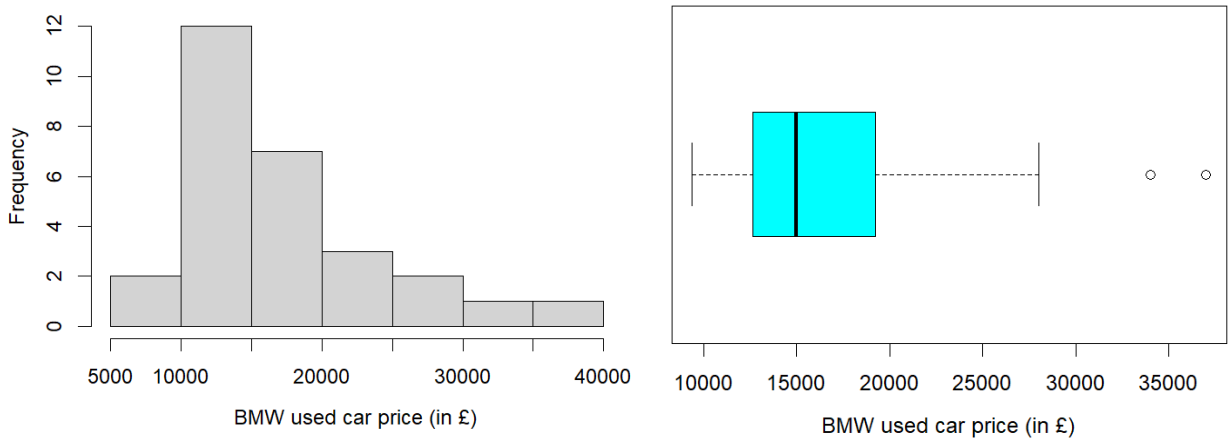
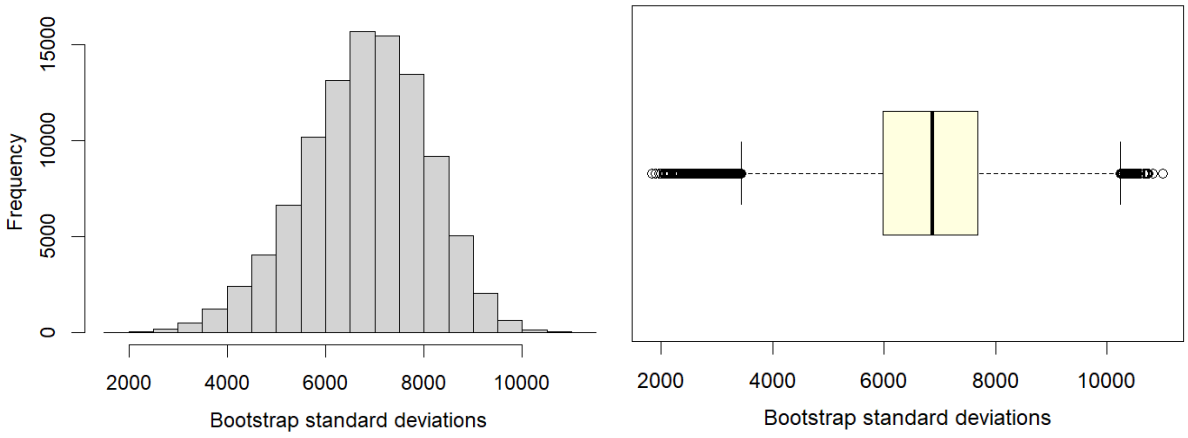


Figure 5: Histogram and boxplot of the BMW used car prices in the United Kingdom (in £) for year 2023 for a sample of 28 sales.

Comments: From histogram and boxplot (Figure 5), distribution is unimodal and is strongly skewed to the right. The boxplot shows two extreme values (outliers) on the right, with the highest selling price of a used BMW car being £36,995, and the smallest price recorded at £9,400. Upon closer examination of the boxplot, it becomes apparent that the quartiles are asymmetric about the median, with the difference between the third quartile and median being larger than the difference between the median and the first quartile. Numerical summaries further support a right-skewed distribution, as the mean is significantly larger than the median.

(b) The empirical (bootstrap) sampling distribution of the sample standard deviation, $\hat{s}$, is shown in Figure 6. The histogram and the boxplot show slightly left skewed.

The bootstrap 99% CI for the population standard deviation is (3328.110, 9622.136).



(i) (ii)

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1842	5989	6877	6803	7691	11008

(iii)

Figure 6: Distribution of bootstrap standard deviations: (i) Histogram; (ii) boxplot (iii) Numerical summary

Under the asymptotic theory, the sample standard deviation is $\hat{s}$=7043.108 and sample size n =28. Then a 99% confidence interval (CI) of the population standard deviation, σ, is

$$\left(\sqrt{\frac{27s^2}{\chi^2_{27,0.005}}},\ \sqrt{\frac{27s^2}{\chi^2_{27,0.995}}}\right)$$

i.e . (5194.082, 10650.393)

(c) The 99% confidence intervals obtained using the non-parametric bootstrap and based on asymptotic theory are considered quite different from each other. As seen in Figure 5, histogram of the sales data does not look normal, and the chi-squared distribution for sample variance is based on the assumption that the raw data are normally distributed, so doesn't look valid. Between the two confidence intervals, the one based on the non-parametric bootstrap would be preferred because it does not rely on additional assumptions, despite the narrower width of the confidence interval based on asymptotic theory.

(d) We use a non-parametric bootstrap to perform a hypothesis test:

H_0: The population mean selling price of a BMW used car is £15,000 or less
H_1: The population mean selling price of a BMW used car is more than £15,000
[or equivalently, H_0: $\mu \leq 15{,}000$ against H_1: $\mu > 15{,}000$]

Firstly, we obtain the sampling distribution of $\bar{x}$ under H_0: $\mu \leq 15{,}000$. Transforming the data set as $x_i^* = x_i - \bar{x} + 15000$, the empirical sampling distribution of $\bar{x}$ under H_0 is given in Figure 7. The vertical line in blue on the graph is the observed mean, $\bar{x}$ =17121.18.

Using *R*, we find *p*-value = 0.05963 (based on 100,000 bootstrap samples). The samples provide some evidence that mean selling price of BMW used car is more than £15,000 at 10% level. However, the claim is not significant at the 5% level.

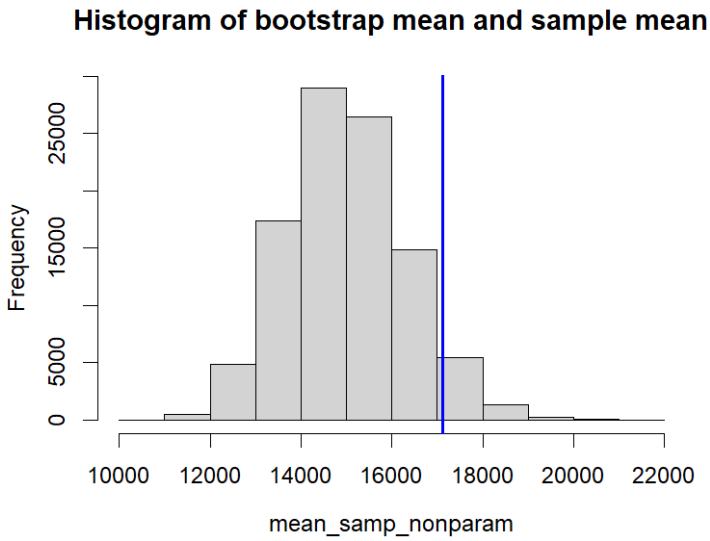


Figure 7: Empirical sampling distribution of mean estimate $\bar{x}$ under $H_0 : \mu \leq 15{,}000$ obtained using non-parametric bootstrap.

(e) We can perform a *t*-test for the hypothesis in 2(d). Using *R* we obtain,

```
t.test(carprice, mu=mean.hyp, alternative="greater")

t = 1.5936, df = 27, p-value = 0.06133
alternative hypothesis: true mean is greater than 15000
```

The *p*-value = 0.06133 is very close to using the non-parametric bootstrap to test and the conclusion is the same. The samples provide some evidence that mean selling price of BMW used car is more than £15,000 at 10% level, but it is not significant at the 5% level.

Comments: In general, *t*-test relies on the assumption that the selling prices data come from a normal distribution, which casts some doubts over its validity here (see Figure 5). Additionally, note that the sample size (n=28) is not particularly large, implying that the Central Limit Theorem (CLT) may not work well in this case. The empirical sampling distribution (Figure 7) shows a slight right skew, but the non-parametric bootstrap test performs well without depending on any assumptions. Thus, of the two methods for testing the population mean, the non-parametric bootstrap may be preferred.

Rubric Presentation

Overall presentation: 8 marks. Follow the rubric in Appendix B. Note that this rubric is written with the intention that 8 marks will only be awarded to an exceptionally good piece of work; I expect most students to get 6 marks; I expect a small number of students to get 4 marks; and I hope only a very few students will get 0 to 3 marks.